EVERY SPECTRAL SPACE EMBEDS AS A CONSTRUCTIBLY CLOSED SUBSET OF A PRODUCT OF TWO-POINT SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of spectral spaces as inverse limits of finite sober spaces at the top of the chapter. Invoked at the start of the counted proof (source lines 4885-4890). Provides the finite quotients used to embed the space.
- The statement that an inverse limit of directed finite sober spaces is spectral. Invoked inside the counted proof (source lines 4895-4898). Identifies the limit of the images as spectral.
- The statement that a closed subset of a spectral space is spectral, and the comparison of limits. Invoked at the end of the counted proof (source lines 4908-4916). Compares the given space with the constructed closed subset.

Source reference. Original lemma lemma-directed-inverse-limit-finite-sober-spectral-spaces: sources/stacks-topology/pinned-original-source.tex, source lines 4851–4851. Navigation pointer only: the counted text refers to the Stacks item lemma-directed-inverse-limit-finite-sober-spectral-spaces, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-fibres-spectral-map-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4589–4589. Navigation pointer only: the counted text refers to the Stacks item lemma-fibres-spectral-map-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-spectral-bijective: sources/stacks-topology/pinned-original-source.tex, source lines 4822–4822. Navigation pointer only: the counted text refers to the Stacks item lemma-spectral-bijective, which is an existing labeled

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Source reference. Original lemma lemma-spectral-sub: sources/stacks-topology/pinned-original-source.tex, source lines 4645–4645. Navigation pointer only: the counted text refers to the Stacks item lemma-spectral-sub, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let W be the topological space with two points, one closed, the other not. A topological space is spectral if and only if it is homeomorphic to a subspace of a product of copies of W which is closed in the constructible topology.*

Proof. Write $W = \{0, 1\}$ where 0 is a specialization of 1 but not vice versa. Let I be a set. The space $\prod_{i \in I} W$ is spectral by Lemma lemma-directed-inverse-limit-finite-sober-spectral-spaces. Thus we see that a subspace of $\prod_{i \in I} W$ closed in the constructible topology is a spectral space by Lemma lemma-spectral-sub.

For the converse, let X be a spectral space. Let $U \subset X$ be a quasi-compact open. Consider the continuous map

$$f_U : X \longrightarrow W$$

which maps every point in U to 1 and every point in $X \setminus U$ to 0. Taking the product of these maps we obtain a continuous map

$$f = \prod f_U : X \longrightarrow \prod\nolimits_U W$$

By construction the map $f : X \to Y$ is spectral. By Lemma lemma-fibres-spectral-map-quasi-compact the image of f is closed in the constructible topology. If $x', x \in X$ are distinct, then since X is sober either x' is not a specialization of x or conversely. In either case (as the quasi-compact opens form a basis for the topology of X) there exists a quasi-compact open $U \subset X$ such that $f_U(x') \neq f_U(x)$. Thus f is injective. Let $Y = f(X)$ endowed with the induced topology. Let $y' \rightsquigarrow y$ be a specialization in Y and say $f(x') = y'$ and $f(x) = y$. Arguing as above we see that $x' \rightsquigarrow x$, since otherwise there is a U such that $x \in U$ and $x' \notin U$, which would imply $f_U(x') \not\rightsquigarrow f_U(x)$. We conclude that $f : X \to Y$ is a homeomorphism by Lemma lemma-spectral-bijective. □